

Giulio Rebuffo

Ethereum Protocol Leader & CTO, Erigon

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Profile

Ethereum protocol engineer and systems programmer holding a leadership role in Ethereum protocol development while leading Erigon, a high-performance Ethereum execution client, and Caplin, Erigon's embedded consensus client. Core experience spans execution-layer architecture, post-Merge consensus integration, p2p protocols, state/history storage, hard-fork implementation, and protocol specification work. Co-author of multiple Final Ethereum Core EIPs included in the Fusaka/Osaka upgrade path, including transaction gas caps, RLP block-size limits, and the CLZ opcode.

Selected Ethereum Protocol Work

- **Final Core EIPs included in Ethereum Fusaka/Osaka:** co-authored EIP-7825, Transaction Gas Limit Cap, EIP-7934, RLP Execution Block Size Limit, and EIP-7939, Count Leading Zeros opcode.
- **Final ecosystem scaling guidance:** co-authored EIP-7935, Set default gas limit to 60M, an informational EIP referenced as an Osaka notice.
- **Ethereum leadership:** active leadership role across Ethereum protocol design, client implementation, and core developer coordination.
- **Economic finality:** contributed to Ethereum economic-finality work spanning proof-of-stake security assumptions, finality behavior, and execution/consensus client implications.
- **Additional authored EIPs:** EIP-7954, Increase Maximum Contract Size. EIP-7975, eth/70 partial block receipt lists. EIP-8178, Binary SSZ Transport for the Engine API. EIP-8202, Scheme-Agile Transactions. EIP-7801, etha Sharded Blocks Subprotocol. EIP-7783 and EIP-7790, controlled gas-limit strategy proposals.
- **Ethereum client architecture:** designed and implemented production systems across execution, consensus, networking, storage, JSON-RPC, Engine API boundaries, and cross-client hard-fork compatibility.

Experience

Erigon Technologies AG

CTO / Project Lead

2021–Present

Zug, Switzerland

- Lead technical direction for Erigon, an Ethereum execution-layer client focused on performance, efficient storage, fast sync, and maintainable protocol architecture.
- Served as a main Erigon engineer for Ethereum's Merge transition, implementing the proof-of-stake transition path and the execution/consensus integration required for post-Merge operation.
- Bootstrapped and led Caplin, Erigon's embedded validating consensus client, enabling Erigon to run as an integrated execution and consensus stack by default.
- Co-authored and drove review for Ethereum Core EIPs now marked Final and included in the Fusaka/Osaka upgrade path: EIP-7825, EIP-7934, and EIP-7939.
- Contributed to hard-fork implementation, testing, and compatibility work across execution payload validation, block limits, transaction limits, precompile/gas changes, p2p behavior, and node configuration.
- Implemented a reinforcement-learning-based p2p peer-selection algorithm for Erigon to improve peer choice and network behavior under production node conditions.
- Implemented and maintained production-grade systems in Go, C++, Rust, and Python, with operational work across Docker, Kubernetes, CI, and large-scale Ethereum test infrastructure.
- Contributed to Silkworm client components in C++ and created an early Rust implementation of Ethereum JSON-RPC in Akula.

Asset Bridging Infrastructure Company

Backend Rust Developer Intern

2018–2020

Remote / undisclosed

- Refactored backend services and internal tooling for an asset-bridging system handling high-value transaction flows.
- Built production experience in Rust, service refactoring, reliability-oriented backend design, and deployment practices for financial infrastructure.

Deepit AG

Backend / Frontend / Blockchain Developer Intern

2017–2018

Genoa, Italy

- Built smart contracts and Ethereum application infrastructure, including Universal Login hooks, POA bridge modifications

for Compound cTokens, ERC-20 deployment tooling, vesting contracts, and wallet/front-end interfaces.

- Built PostgreSQL-backed indexing and transaction-tracking systems. Worked across Solidity, TypeScript, JavaScript, WebAssembly/Emscripten, React, Angular, Truffle, and Hardhat.

University of Genoa

Embedded Systems / Robotics Research Assistant

2016–2017

Genoa, Italy

- Supported robotics research tasks using ROS, C++, and Python after independently building embedded and robotics projects.
- Built a Unity/C# drone path-finding simulation and experimented with speech-recognition interfaces for robotics workflows.

Technical Skills

Languages: Go, C, C++, Rust, Python, Solidity, TypeScript, JavaScript, C#

Ethereum / Protocols: Execution clients, consensus clients, Engine API, JSON-RPC, DevP2P, LibP2P, Beacon API, EVM, RLP, SSZ, gRPC, REST

Storage / Infrastructure: MDBX, LMDB, PostgreSQL, MongoDB, Docker, Kubernetes, Linux, CI, distributed test infrastructure

Education

Università della Svizzera italiana / University of Lugano

B.S. Computer Science & Engineering

2022–Present

Lugano, Switzerland

The American School in Switzerland

AP Diploma, Highest Honors

2018–2022

Lugano, Switzerland

- SAT: 1480. AP scores: Calculus AB 5, Calculus BC 5, Computer Science 5, Macroeconomics 5, Statistics 5, Italian 5.